

Body-fitted external flow validation

SIMPLE/Rhie-Chow; first order upwind; strict independent physical error <3%.

NACA reference: vector-digitized Di Ilio 2020, uncertainty +/-0.0001.

Historic NACA 0.205/0.120 pass claim is withdrawn.

Single grid and residual convergence do not establish physical accuracy.

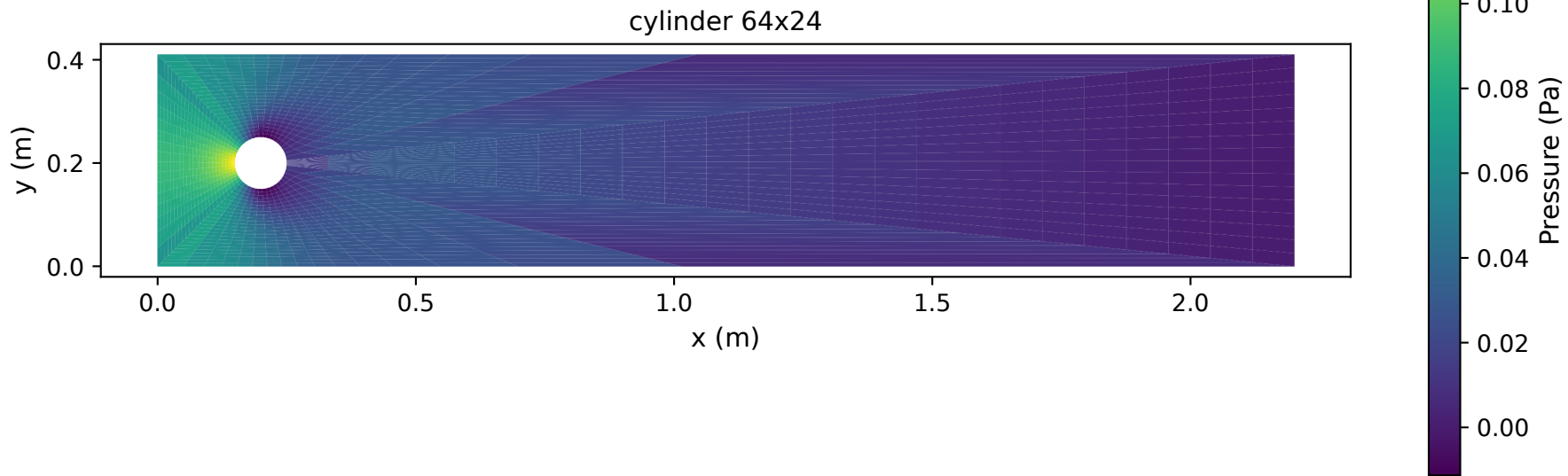
cylinder 64x24 cpu: converged=True, passed=False

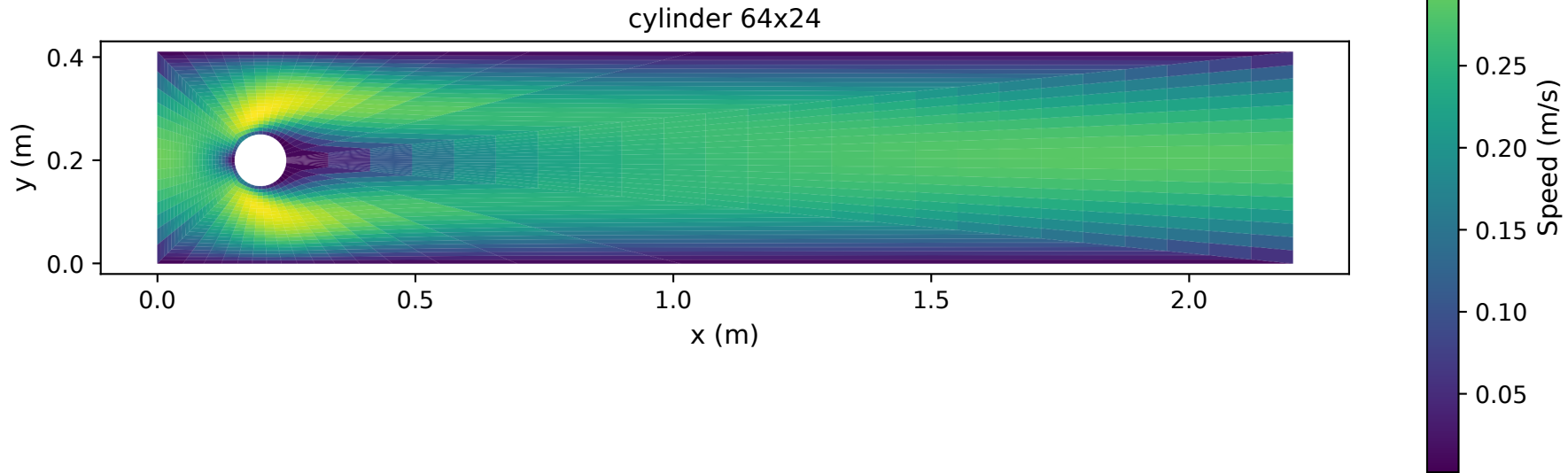
{'drag': 6.324792432227089, 'lift': 0.015101732296668986, 'pressure_difference_pa': 0.13099149005493882}

{'drag_relative_error': 0.13356976291987346, 'lift_relative_error': 0.4221495471147568,

'pressure_difference_pa_relative_error': 0.11462988380860376, 'solver_residual': 9.95675123522303e-07,

'convergence_failure': 0}





Wall pressure (Pa)

